Henry Chen

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https://github.com/Chenry513

Education

Master of Science in Applied Computing (CGPA: 91%) — Sept. 2025 – Dec. 2026 (est.)
British Columbia Institute of Technology — Burnaby, BC

Bachelor of Science in Computing Science — Sept. 2020 – Aug. 2025
Simon Fraser University — Burnaby, BC

Technical Skills

Programming Languages: Python, C/C++, SQL, JavaScript, TypeScript
Frameworks & Technologies: Node.js, Express, REST APIs, Three.js, Pandas, NumPy, OpenCV
Databases & Tools: MongoDB, MySQL, PostgreSQL, SQLite, Git, GitHub Actions, Docker, Docker Compose, Linux, npm

Work Experience

Mitacs Research Intern – Robotics & Computer Vision — Jan. 2026 – Present
BCIT — Seaspan Shipyards Collaboration (Supervisor: Mathew Smith) — Vancouver, BC

- Developing a Python-based data pipeline to collect and label welding image data for computer vision training.
- Building and evaluating machine learning models to detect welding joints for robotic automation.
- Collaborating with mechanical and robotics engineers to integrate vision models into real-world welding systems.

Research Assistant (Backend & Automation Engineer) — May – Aug. 2025
Simon Fraser University — PadComputing Lab (Prof. Nicholas Vincent) — Burnaby, BC

- Designed and implemented a GitHub-integrated backend ingestion pipeline to automate validation and integration of structured user submissions, reducing manual Django-admin update time by 60%.
- Implemented a Python-based linter using schema validation to enforce Django model constraints.
- Integrated GitHub Actions CI to automatically validate and process submissions on each commit.
- Prevented invalid entries from being deployed by enforcing schema compliance before publication.

Research Assistant (Backend Engineer) — Aug. – Dec. 2024
Simon Fraser University — PadComputing Lab (Prof. Nicholas Vincent) — Burnaby, BC

- Engineered backend components for a Django-based research platform supporting structured LLM metadata and user-facing functionality.
- Designed relational models and schema logic for structured metadata storage and preference management.
- Built 8+ RESTful API endpoints using Django REST Framework for authentication and data retrieval.
- Enhanced Django Admin and managed migrations to maintain data integrity across updates.

Projects

Applied Data Science, BCIT — Dec. 2025
30-Day Hospital Readmission Prediction — github.com/Chenry513/COMP-9170-Final-Project

- Developed a patient-level 30-day hospital readmission risk model using 100K+ clinical encounters while preventing data leakage through GroupKFold cross-validation.
- Built a reproducible ML pipeline with imbalance handling and leakage-free stacking for reliable model evaluation.
- Evaluated performance using ROC-AUC, PR-AUC, and bootstrap confidence intervals to quantify uncertainty and prevent misleading performance claims in a healthcare setting.

Affective Computing, SFU — Apr. 2025
Emotion-Based Break Recommendation System — github.com/Chenry513/CMPT419_Final_project

- Built and evaluated two temporal deep learning pipelines (OpenFace+LSTM and CNN+LSTM) to detect frustration from gameplay video.
- Modeled emotional dynamics over time rather than relying on frame-level classification.

- Collected and manually annotated video clips (Anger vs Neutral), achieving 70% accuracy and F1 = 0.70.
- Deployed the trained model via a Flask web application enabling real-time video uploads and automated break recommendations.

Real-Time Web Application Personal Project July 2024
Rankr github.com/Chenry513/rankr

- Built a real-time polling system enabling groups to create, join, and rank shared options collaboratively.
- Implemented live updates using React, TypeScript, Socket.io, and Redis for low-latency synchronization.
- Designed backend logic to aggregate ranked votes and compute final group decisions.

Computer Vision, SFU May 2023
CIFAR-100 Image Classification github.com/Chenry513/PyTorch-Image-Classifier

- Designed and trained a convolutional neural network on the CIFAR-100 dataset (100 classes, 60K images), achieving competitive validation performance.
- Ranked 13th out of 120 students in a graduate-level computer vision Kaggle competition.
- Improved generalization using data augmentation and regularization to reduce overfitting.